4.3. Angiotensin II Receptor Blockers

The renin–angiotensin system (RAS) plays an integral role in blood pressure regulation and fluid homeostasis, and its hyperactivity has been classically linked to hypertension. However, growing evidence suggests a pathological role of RAS in RA, particularly through angiotensin-converting enzyme (ACE), which is overexpressed in the peripheral blood and synovium of RA patients [Wang2013].

Angiotensin II stimulates angiotensin II type 1 receptors (AT1R), triggering the generation of reactive oxygen species (ROS) and activation of nuclear factor kappa B (NF-κB), which subsequently promotes the release of multiple pro-inflammatory cytokines [Mostafa2020]. In contrast, angiotensin II type 2 receptors (AT2R) counterbalance AT1R, and their stimulation has been linked to anti-inflammatory and tissue-protective effects [Wang2013].

In a preclinical study, adjuvant-induced arthritis (AIA) rats treated with losartan at doses of 10 and 15 mg/kg had a significant reduction in their polyarthritis index compared with vehicle-treated rats, although this improvement was smaller than that seen with methotrexate (MTX) [Wang2013]. Losartan also shifted the balance between the two receptors, with less AT1R and more AT2R in the spleen and synovium, and this shift tracked the fall in the polyarthritis index [Wang2013]. Another study on AIA rats found that losartan lowered serum IL-6 and TNF-α compared with the untreated arthritic group, together with a rise in serum glutathione (GSH) and a fall in serum malondialdehyde (MDA) [AbdelElGaphar2018]. An in vitro study tested losartan, valsartan, and enalapril on peripheral blood mononuclear cells (PBMCs) collected from 16 RA patients. Only losartan clearly lowered four cytokines (IL-6, IL-17F, IL-22, and IFN-γ), while valsartan and enalapril did little, which points to an effect of losartan itself rather than of the whole class [Cardoso2018].

The main clinical evidence for this class in RA comes from a trial of another ARB, candesartan. In this Egyptian single-blind, randomized, placebo-controlled trial, 45 patients with RA (15 per group) were treated for 3 months with add-on candesartan (8 mg/day), atorvastatin (20 mg/day), or placebo [Mostafa2020]. Candesartan significantly lowered serum IL-1β, TNF-α, CRP, and erythrocyte sedimentation rate (ESR), and it improved DAS28 and HAQ-DI disability scores compared with both baseline and the control group [Mostafa2020]. However, unlike the antioxidant effect of losartan in rats, candesartan failed to produce significant changes in MDA and rheumatoid factor (RF), whereas atorvastatin significantly lowered both markers [Mostafa2020]. The interpretation of these findings is further limited by baseline imbalances between the groups in several clinical variables and by the small sample size [Mostafa2020].

ARBs, particularly losartan, may represent a reasonable antihypertensive option for patients with RA and hypertension; however, larger and longer-term randomized controlled trials, especially those evaluating losartan, are needed to confirm these findings.

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